Operating Systems – Final Examination

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Total: 100 points

1. Process Scheduling:
   (a) (5 points) Describe two general approaches to load balancing of multiple-processor scheduling.
   (b) (10 points) What is Little's formula? Explain why scheduling analyses based on it may be questionable.

2. Synchronization:
   (a) (10 points) The first known correct software solution to the critical-section problem for two processes was developed by Dekker. The two processes, P_0 and P_1, share the following variables:

```
boolean flag[2];  /* initially false */
int turn;
```

The structure of process P_i (i == 0 or 1) is shown below; the other process is P_j (j == 1 or 0). Prove that the algorithm satisfies all three requirements for the critical-section problem.

```
do {
  flag[i] = true;

  while (flag[j]) {
    if (turn == j) {
      flag[i] = false;
      while (turn == j)
        ; /* do nothing */
      flag[i] = true;
    }
  }

    /* critical section */

  turn = j;
  flag[i] = false;

    /* remainder section */
} while (true);
```

   (b) (10 points) Regarding monitor, what are the two common signal models of condition variables? Explain the difference between them.

3. Deadlocks:

(a) (10 points) Explain the four necessary conditions of a deadlock.

(b) (5 points) Explain why an unsafe state MAY BUT NOT NECESSARILY lead to a deadlock.

(c) (10 points) Consider the following snapshot of a SAFE system; can a request (0,4,2,0) from p_1 be granted? Show your answer using the banker's algorithm.

	Allocation				Max			
	A	*B*	*C*	*D*	*A*	*B*	*C*	*D*
P_0	0	0	1	2	0	0	1	2
P_1	1	0	0	0	1	7	5	0
P_2	1	3	5	4	2	3	5	6
P_3	0	6	3	2	0	6	5	2
P_4	0	0	1	4	0	6	5	6

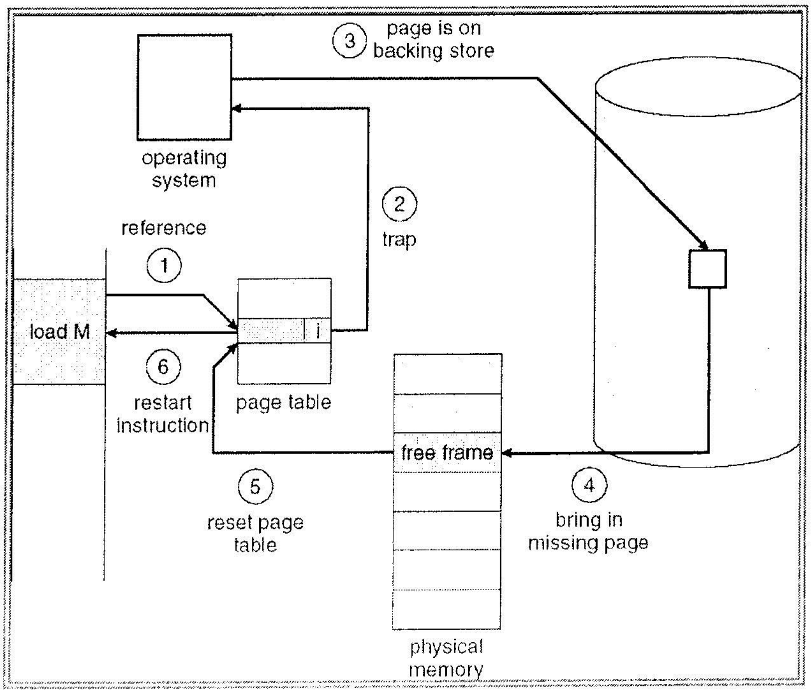
Available			
A	*B*	*C*	*D*
1	5	2	0

4. Address Binding — (10 points) Technically, the binding of instructions and data to physical memory addresses can be done at *compile time*, *load time*, or *execution time*. Explain the three binding approaches and show the weakness of the first two.

5. Virtual Memory:

(a) (5 points) Show two advantages of virtual memory over paging.

(b) (10 points) The following figure shows the procedure for handling a page fault. Explain the function of steps 1 to 6. Here, we assume a free frame is always available.



6. Paging — Consider a paging system that supports 4-bit logical address and has 32 bytes physical memory. The size of a page is 4 bytes.

(a) (5 points) What is the size of the page table if each table entry costs 2 bytes (the first byte for frame number & the second byte for protection bits)?

(b) Suppose the system employs the technique of **TWO-LEVEL PAGING** and registers are used to store the outer page table. Allocate memory (i.e., frames) for the following process and the corresponding page table. Note: (i) You first allocate memory for the page table, and then the process. (ii) Free frames are stored in a queue. Frame 5 is the first free frame.